

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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Anonymous submission

ABSTRACT

Representation alignment (REPA) accelerates generative training by distilling representations from a strong, pretrained vision encoder into the intermediate features of a diffusion transformer. This work resolves a fundamental question about the mechanism underlying that gain: which property of the target representation actually governs generation, its *global* semantic content (e.g., as measured by ImageNet-1K accuracy) or its *spatial* structure (i.e., the pairwise cosine similarity among patch tokens)? The prevailing assumption holds that stronger global semantic performance yields a better alignment target. Through a large-scale empirical analysis spanning 27 vision encoders and three model scales, we establish the opposite: it is spatial structure, rather than global performance, that determines the generative utility of a target representation. To characterize this dependence quantitatively, we introduce a family of fast-to-compute spatial self-similarity metrics whose values correlate far more strongly with generation quality than does linear probing accuracy. Building directly on this finding, we introduce two minimal modifications (<4 lines of code) that explicitly accentuate the transfer of *spatial* information: we replace the standard MLP projection layer in REPA with a single convolutional layer, and apply a spatial normalization layer to the external representation. The resulting method, termed iREPA, consistently accelerates the convergence of REPA across a diverse range of vision encoders, model sizes, and training recipes—including REPA, REPA-E, MeanFlow, and JiT. Together, these results reframe the working mechanism of representation alignment and chart a concrete path toward exploiting it for more efficient training of generative models.

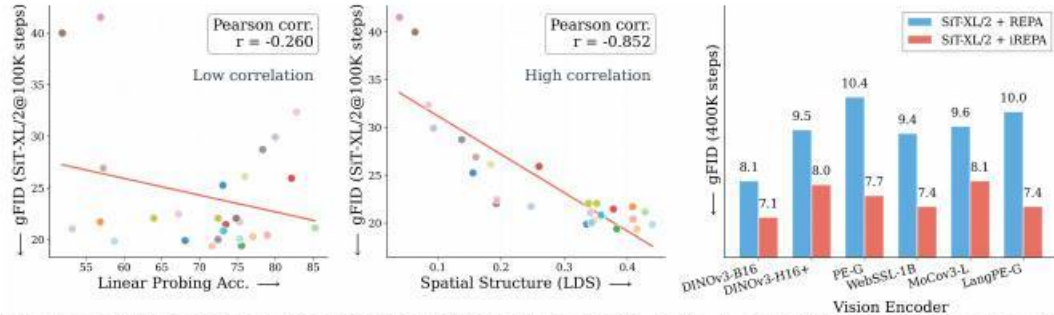


Figure 1: **What matters for representation alignment?** *Left:* A correlation analysis across 27 diverse vision encoders. Contrary to the prevailing assumption, it is the spatial structure of a target representation—rather than its global performance, as measured by linear probing accuracy—that governs generation performance. *Right:* Building on this finding, we introduce two minimal modifications that accentuate the transfer of spatial features from the target representation to the diffusion model; the resulting method consistently accelerates the convergence of REPA across diverse settings.

1 INTRODUCTION

Representation alignment has emerged as one of the most effective techniques for accelerating the training of diffusion transformers (Ma et al., 2024; Peebles & Xie, 2023). By aligning the internal



Figure 2: **Motivating examples — spatial structure matters.** These comparisons reveal a clear inversion between ImageNet accuracy and generation quality. *Left:* PE-G, despite far higher validation accuracy (82.8% vs. 53.1%), yields worse generation than SpatialPE-B (Bolya et al., 2025). *Right:* WebSSL-1B (Fan et al., 2025) likewise attains much higher global performance (76.0% vs. 53.1%) yet worse generation. **Spatial Self-Similarity:** Spatial structure, by contrast, is the reliable predictor of generation quality; see §3 for the spatial structure metric. All results reported at 100K iterations using SiT-XL/2 and REPA.

representations of a diffusion transformer with those of a pretrained self-supervised visual encoder, recent methods deliver substantial gains in both convergence speed and final generation quality (Yu et al., 2024; Leng et al., 2025a). Yet despite these empirical successes, the precise mechanism through which self-supervised features enhance diffusion training has remained poorly understood. This gap leaves open a fundamental question: does the improvement arise principally from richer global semantic information, as commonly quantified through linear probing accuracy, or from the faithful transfer of spatial structure, characterized by the relationships among patch token representations?

Resolving this question is central to advancing generative model training, since it directly determines how practitioners should select a target representation and extract its full benefit. The prevailing assumption holds that an encoder’s suitability for representation alignment tracks its ImageNet-1K validation accuracy—a proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). Under this view, target representations with higher ImageNet accuracy are expected to yield better generation (Yu et al., 2024). Consistent with it, gains in the linear probing accuracy of diffusion features are routinely invoked as evidence for the effectiveness of representation alignment, further entrenching global semantic information as the presumed primary driver of improvement. The following passage from REPA (Yu et al., 2024) crystallizes this conventional understanding:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

In this paper, we overturn this conventional wisdom through a systematic investigation of what truly drives representation alignment: global semantic information or spatial structure. Conducting a large-scale empirical analysis across a broad and diverse set of vision encoders—including recent large-scale vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perception encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024)—we arrive at three findings that fundamentally reshape this understanding.

Higher validation accuracy does not imply a better representation for generation. Contrary to the prevailing assumption, vision encoders with stronger global semantic performance, as measured by ImageNet-1K linear probing accuracy, frequently *underperform* as alignment targets. Consider PE-Spatial-B, a spatially tuned model derived from PE-Core-G (Bolya et al., 2025). Although PE-Spatial-B attains far lower patch-token validation accuracy¹ than PE-Core-G (53.1% vs. 82.8%), it yields markedly better generation with REPA (Figure 2). WebSSL-1B (Fan et al., 2025) exhibits the same inversion: substantially higher global accuracy (76.0% vs. 53.1%), yet worse generation. As we show in §2, this pattern recurs consistently across a wide range of target representations, pointing to a fundamental principle governing how representation alignment benefits diffusion training.

Spatial structure, not global information, determines generation performance. To make this insight quantitative, we introduce a family of straightforward metrics that measure the spatial

¹Following REPA (Yu et al., 2024), we measure the global semantic performance of an external representation via linear probing accuracy on patch tokens, since only patch tokens are used for representation alignment.

self-similarity structure (Shechtman & Irani, 2007) among patch tokens (§3). We then conduct a large-scale correlation analysis relating REPA generation FID to linear probing accuracy and to spatial structure across 27 vision encoders and three model scales (SiT-B, SiT-L, SiT-XL). The result is decisive: every spatial structure metric correlates strongly with generation FID (Pearson $|r| > 0.852$), far surpassing the predictive power of ImageNet-1K validation accuracy ($|r| = 0.26$). Complementary controlled experiments reinforce the conclusion, demonstrating that representations carrying very little global information can nonetheless deliver substantial REPA gains. SAM2, for instance, yields better generation than encoders with roughly 60% higher ImageNet accuracy (Fig. 3); and even classical, purely spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005)—though not the strongest targets—produce meaningful gains under representation alignment (§3).

Accentuating spatial features improves convergence. Motivated by these findings, we introduce two minimal modifications (<4 lines of code) that explicitly amplify the transfer of *spatial* information from the target representation to the diffusion model. First, we add a spatial regularization layer that sharpens the spatial contrast of the target representations. Second, we identify the standard MLP projection layer—used to map diffusion features onto the target representation dimensionality—as a source of spatial information loss (Figure 6a), and replace it with a single convolutional layer that preserves local spatial relationships by construction. The effect is striking and consistent: the resulting method, iREPA, reliably accelerates the convergence of REPA across diverse encoders, model sizes, and training recipes, including REPA-E (Leng et al., 2025a) and MeanFlow (Geng et al., 2025) with REPA.

We summarize the principal contributions of this paper as follows:

- **A mechanistic account of representation alignment.** Through a large-scale empirical analysis across 27 vision encoders and three model scales, we establish that spatial structure—not global semantic information—is the property that drives the effectiveness of representation alignment.
- **A predictive spatial structure metric.** We introduce a straightforward, fast-to-compute Spatial Structure Metric (SSM) that correlates substantially more strongly with downstream FID than linear probing accuracy, and that explains generation trends left unexplained by prior work.
- **iREPA, a minimal and general recipe.** We translate this mechanistic insight into two simple modifications that accentuate the transfer of spatial information from the target representation to the diffusion features. The resulting method delivers consistent convergence-speed improvements over REPA across variations in target representation, model architecture, and training recipe (REPA, REPA-E, MeanFlow with REPA, and JiT with REPA).

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin by presenting a series of motivating observations that jointly establish a clear conclusion: higher global performance (ImageNet-1K accuracy) of a target representation does not translate into better generation with REPA. In §3 we show that these observations, which resist explanation under the conventional view, are precisely accounted for once the spatial structure of the target representation is measured.

Recent vision encoders. We first examine a range of recent vision encoders, including Perception Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a compact spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). As Figure 2 shows, PE-Core-G attains far higher ImageNet-1K accuracy (82.8% vs. 53.1%) yet performs distinctly worse as the REPA target representation (FID 32.3 vs. 21.0). WebSSL-1B (1.2B) exhibits the identical inversion: despite substantially higher ImageNet-1K accuracy (76.0% vs. 53.1%), it yields worse generation when used as the target representation for REPA (FID 26.1 vs. 21.0).

SAM2 outperforms vision encoders with far higher ImageNet-1K accuracy. To probe how little global information a target representation actually requires, we analyze the SAM2-S vision encoder (46M) (Ravi et al., 2024)—a small model carrying very little global information, with a validation accuracy of only 24.1% (Fig. 3a). Despite this, when used for REPA, SAM2-S achieves a better FID than vision encoders with dramatically higher ImageNet-1K accuracy, PE-Core-G (82.8%) among them.

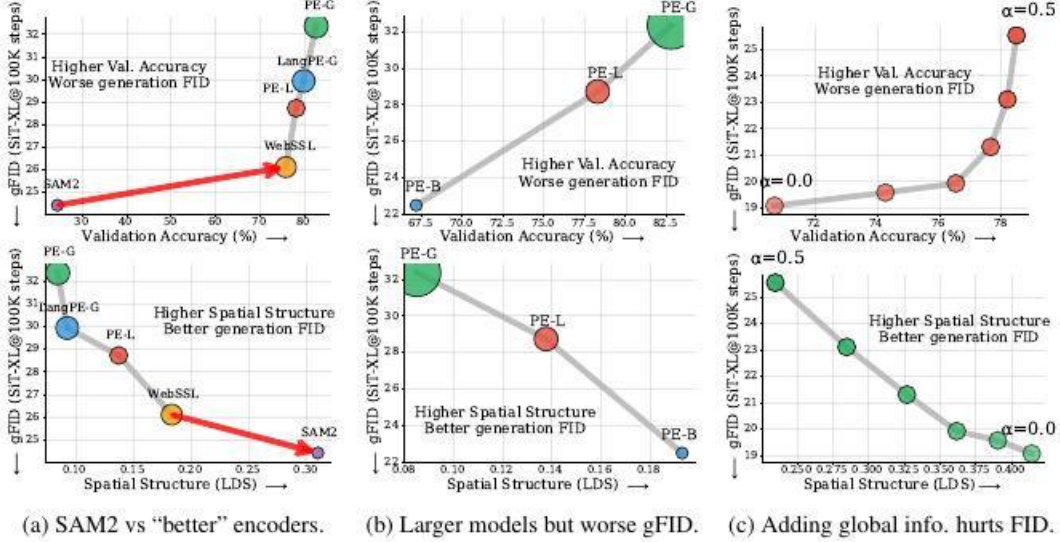


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** Across several distinct settings, global performance fails to track generation FID under REPA. (a) SAM2-S, with a validation accuracy of only 24.7%, achieves better generation with REPA than models with $\sim 60\%$ higher validation accuracy. (b) Larger encoders within the same family can attain higher validation accuracy yet worse generation. (c) Injecting global information into patch tokens via the CLS token raises global performance but degrades generation. **Bottom row:** Spatial structure, rather than global performance, is the reliable indicator of generation quality; see §3 for the large-scale analysis across spatial structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

Larger models within the same encoder family can yield similar or worse generation. It is commonly assumed that larger models within an encoder family carry better representations, as reflected in ImageNet-1K accuracy. For representation alignment, however, the larger variants often produce comparable (DINOv2) or even *worse* (PE, C-RADIO) generation performance (Fig. 3b). Notably, (Yu et al., 2024) report a similar observation for DINOv2, attributing it to the conjecture that “all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations.” We later show that spatial structure offers a more principled explanation of this trend (§3).

Injecting more global information can hurt generation. To isolate the effect of global information, we conduct a controlled experiment that deliberately injects global semantics—via the CLS token—into the local patch tokens (using DINOv2 as the encoder). The CLS-token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ modulates the mixing strength. As Figure 3c shows, increasing α from 0 to 0.5 raises linear probing accuracy monotonically from 70.7% to 78.5%; yet generation quality degrades in lockstep, with FID worsening from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$. Thus, adding global information improves the very proxy that is presumed to predict REPA quality while actively harming the outcome it means to predict.

Taken together, these observations demonstrate that the global performance of a representation is not a reliable indicator of its REPA effectiveness. We show next (§3) that spatial structure supplies exactly the predictive signal that global performance lacks.

- **Finding 1.** Stronger global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation; across encoders, families, and controlled interventions, global performance fails to predict REPA effectiveness.

3 SPATIAL STRUCTURE MATTERS MORE

We advance and rigorously test a single hypothesis: it is the spatial structure of a target representation, rather than its global information, that governs generation performance. To this end, we first formalize a family of straightforward metrics that quantify the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. We then establish two results: every spatial structure metric

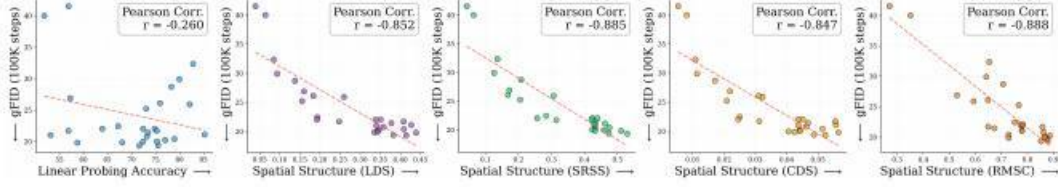


Figure 4: Spatial structure correlates far more strongly with generation quality than linear probing. Correlation analysis across 27 diverse vision encoders using SiT-XL/2 and REPA. Linear probing correlates only weakly with FID (Pearson $|r| = 0.260$), whereas every spatial structure metric—LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$)—correlates substantially more strongly with generation performance. See Fig. 10 for detailed plots with encoder labels.

correlates far more strongly with gFID than ImageNet-1K accuracy does, and these same metrics account for the previously unexplained observations of §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define:

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ be the extract patch representations, with $T = H \times W$ patches. Let kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{(\mathbf{x}_t, \mathbf{x}_{t'})}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ be the spatial location of each of the T tokens in coordinate space and $d(\cdot, \cdot) \in \mathbb{N}$ be the Manhattan distance between pairs of tokens.

We then measure *spatial* self-similarity metric as how self-similarity $K_{\mathcal{X}}$ varies with lattice distance d between patch tokens. Intuitively, larger values indicate stronger spatial organization (closer patches more similar to each other than patches further away). By default, we use a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is computed as the expectation over patch representation $\mathcal{X}$. We use $r_{\text{near}} = r_{\text{far}} = H/2$ here, though we found correlation to be robust to their exact choices. Please see Appendix B for exact details and alternative spatial metrics (CDS, SRSS, RMSC) which also perform effectively (see Fig. 4).

Spatial structure correlates much higher with generation performance than linear probing. We next perform large-scale correlation analysis across 27 diverse vision encoders. As shown in Figure 4, we find that while typically used linear probing shows very weak correlation (Pearson $|r| = 0.26$), all SSM metrics show much higher correlation with generation performance (Pearson $|r| > 0.85$).

Generalization across model scales. We verify the correlation across different model scales (SiT-B, SiT-L, SiT-XL) in Figure 5. Linear probing shows weak correlation across model scales ($|r| < 0.306$), while spatial structure shows much higher correlation with generation performance ($|r| > 0.826$).

- **Finding 2.** *Spatial structure correlates much higher with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends.*

Spatial structure can explain previously unexplained trends. As discussed in §2, global performance (ImageNet-1K accuracy) does not serve as a predictive measure of effectiveness for representation alignment. We find that instead the above spatial metrics serve as better predictors.

(1) *PE-Spatial-B vs PE-Core-G:* Figure 2 shows that PE-Core-G achieves much higher ImageNet-1K accuracy (82.8% vs 53.1%), but performs worse when used as target representation for REPA (FID 32.3 vs 22.0). Looking at spatial structure metric makes things clearer. As seen in Figure 2, despite lower global performance, PE-Spatial-B shows much better spatial structure than PE-Core-G; leading to better generation performance as observed.

(2) *SAM2 outperforms “better” vision encoders:* §2 shows that while SAM2 achieves much lower ImageNet-1K accuracy (24.1%), it leads to better generation than encoders with $\sim 60\%$ higher accuracy. As in Fig. 3a; these gains can be directly explained through SAM2’s better spatial structure.

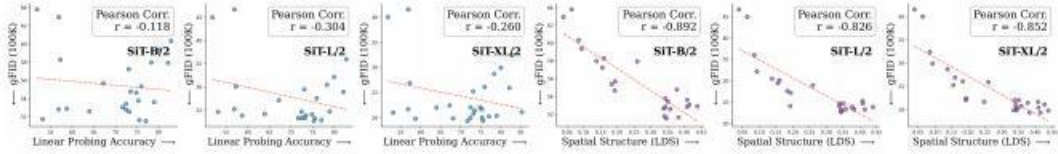
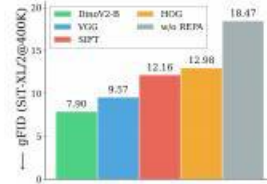


Figure 5: **Correlation analysis across model scales.** At every model scale, spatial structure (right) correlates consistently more strongly with gFID than linear probing (left), confirming that the predictive advantage of spatial structure is scale-invariant.

(3) *Larger models in same encoder family underperform:* As shown in Fig. 3b, while larger models in same encoder family show better ImageNet-1K accuracy, they can have worse spatial structure, leading to worse generation performance with REPA. This also aligns with observations from Yu et al. (2024), where they find that larger models can have similar or worse generation performance.

(4) *Adding global information to patch tokens via CLS token hurts generation:* In Figure 3c, we observe that increasing global information by mixing the CLS token with patch tokens improves global performance. However, mixing CLS token reduces spatial contrast among patch tokens. This leads to patch tokens having high similarity with otherwise unrelated tokens (e.g., from foreground object to background). The reduced spatial structure thus leads to worse generation performance.

If spatial structure matters more, can we use SIFT or HOG features for REPA? Surprisingly, yes. We find that while certainly not the best, classical spatial features like SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005) and intermediate VGG features (Simonyan & Zisserman, 2014) all lead to performance gains with REPA. This provides further evidence that representation alignment can benefit from spatial features alone without need for additional global information.



Can we use spatial metrics to explain gains with REPA? Yes. As shown, the introduced spatial metrics can be used to explain both gains with REPA as well as our improved training recipe (iREPA) which we introduce next in §4.



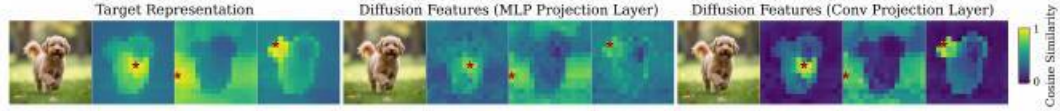
4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

Having established that spatial structure governs the generative utility of a target representation, we now operationalize this principle. We introduce two minimal, architecturally targeted modifications to the original REPA training recipe, each designed to accentuate the transfer of spatial structure from the teacher (the pretrained vision encoder) to the student (the diffusion transformer). Together they constitute iREPA, a recipe that requires fewer than four lines of code yet consistently improves convergence.

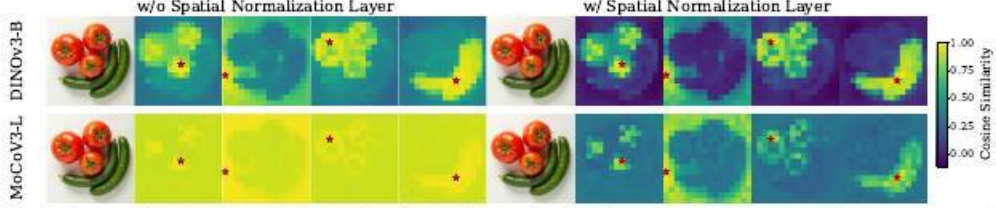
Convolutional projection layer instead of MLP. Standard REPA employs a three-layer MLP projection to map the diffusion feature dimensionality onto that of the external representation. Because this projection acts on each patch token independently, it discards the spatial relationships among tokens: as Figure 6a demonstrates, the MLP mapping is lossy and measurably attenuates the spatial contrast between patch tokens. We therefore replace it with a lightweight convolutional projection (kernel size 3, padding 1) that operates directly over the spatial grid. By construction, the locality of the convolutional operator imposes an inductive bias that preserves the neighborhood relationships among patch tokens, precisely the property our analysis identifies as decisive for generation.

Spatial normalization layer. Consistent with Siméoni et al. (2025), we find that the patch tokens of pretrained vision encoders carry a substantial global component, evidenced by the high linear probing accuracy obtained from the mean of the patch tokens (Figure 14). Figure 6b makes the trade-off explicit: mixing this global component (the mean of the patch tokens) into the individual tokens raises global performance, but at the cost of the spatial contrast between tokens, so that unrelated tokens (e.g., a foreground object and the background) come to appear spuriously similar.

The mechanistic account of §3 yields a direct prescription: deliberately suppressing this global component should sharpen the spatial contrast among patch tokens and thereby strengthen the



(a) *Simpler projection layer for REPA.* The standard MLP projection layer in REPA (middle) discards spatial information when transferring features from the target representation (left) to the diffusion features. Replacing it with a simpler convolutional layer preserves this spatial information far more faithfully (right).



(b) *Spatial normalization layer.* The patch tokens of pretrained vision encoders carry a global component that suppresses spatial contrast, causing tokens in one semantic region (e.g., tomato) to exhibit substantial cosine similarity with unrelated tokens (e.g., background or cucumber). Suppressing this global component sharpens the spatial contrast among patch tokens and thereby improves generation performance.



(c) *Overall impact of the improved training recipe (iREPA) on the spatial structure of the diffusion features.*

Figure 6: Motivation for the two minimal modifications (iREPA) that accentuate the transfer of spatial features from the target representation to the diffusion features. All results reported with SiTL/2 and REPA at 100K iterations.

pairwise-similarity signal that drives REPA. We therefore apply a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ denotes the patch tokens, the expectation and variance are computed across the spatial dimension, γ controls the fraction of the global component that is removed, and $\epsilon = 10^{-6}$ ensures numerical stability.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```

4.1 EXPERIMENTS

We validate iREPA through an extensive suite of experiments on ImageNet 256×256 , organized around three research questions that together establish its effectiveness, scalability, and generality:

- Does iREPA consistently accelerate the convergence of diffusion transformers over REPA across a diverse set of external representations? (Figure 7, 8, 12, 13, 15)
- Does iREPA scale with model size and generalize across variations in training configuration? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Does iREPA generalize to more recent representation alignment methods such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025), and to pixel-space diffusion models such as JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence Speed. We evaluate the convergence behavior of iREPA across a diverse set of vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and two model sizes

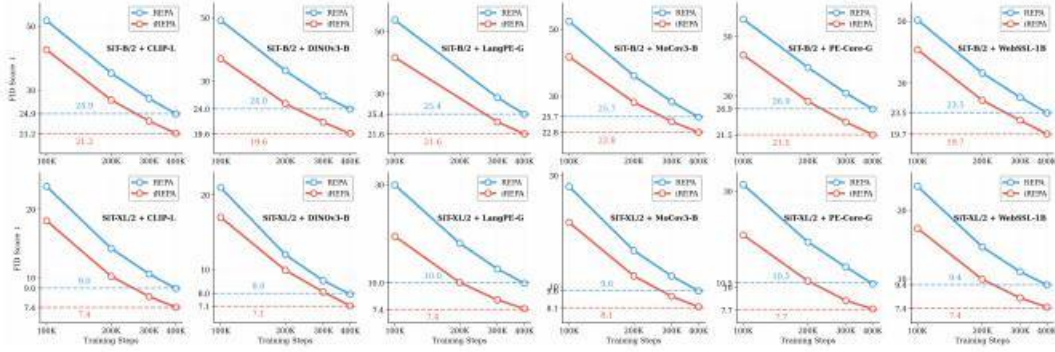


Figure 7: **Accelerating spatial features consistently improves convergence speed.** FID convergence curves show that iREPA accelerates convergence over baseline REPA across every vision encoder and model size considered.

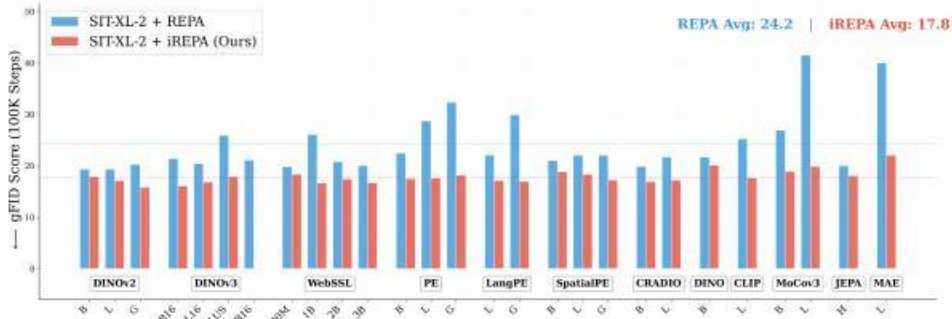


Figure 8: **Variation in target representation.** Across all 27 vision encoders, accentuating the transfer of spatial features from the target representation to the diffusion features (iREPA) yields consistent improvements in convergence speed. See Appendix C for further results across diverse settings.

(SiT-XL/2 and SiT-B/2), reported in Fig. 7. Across every combination of target representation and model size, iREPA attains faster convergence than baseline REPA, establishing that the benefit of accentuating spatial transfer is uniform rather than encoder- or scale-specific.

Target representation. We assess the generality of iREPA across vision encoders in Fig. 8 and Table 4. iREPA improves generation quality for every vision encoder considered, with the full comparison across all 27 encoders reported in Appendix C, E.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60	SiT-B	27.5	49.50	7.00	0.46	0.59	Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62	+iREPA	34.1	43.37	7.87	0.50	0.60	+iREPA	41.7	36.0	6.61	0.52	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60	SiT-L	55.7	24.10	6.25	0.62	0.60	Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	76.3	17.6	6.36	0.65	0.61	+iREPA	66.9	20.28	6.14	0.63	0.62	+iREPA	42.4	35.3	6.90	0.53	0.62
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59	SiT-XL	70.3	19.06	5.83	0.65	0.61	Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	70.8	19.5	6.35	0.64	0.61	+iREPA	77.9	16.96	6.26	0.66	0.61	+iREPA	38.3	38.9	7.19	0.51	0.62

(a) Vision Encoder size

(b) Model size

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** iREPA delivers consistent gains over baseline REPA across encoder size (a), diffusion model size (b), and alignment depth (c). Unless otherwise specified, all results use DINOv2 as the encoder, SiT-XL/2, 100K steps, and vanilla REPA as the baseline.

Encoder size. Table 1a reports how iREPA generalizes across encoder sizes within the PE family, spanning PE-B (90M), PE-L (320M), and PE-G (1.88B). iREPA improves performance at every encoder size, and, notably, the relative improvement grows monotonically with encoder capacity

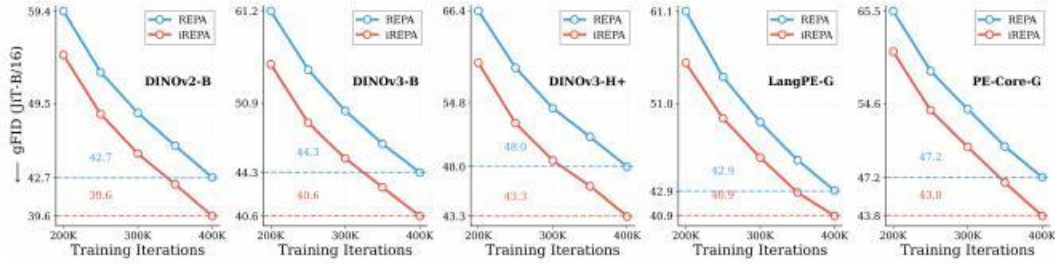


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves comparing REPA and iREPA with pixel-space JiT (Li & He, 2025b) across vision encoders. Accentuating the transfer of spatial structure again yields consistently faster convergence across encoders, demonstrating that the principle carries over to pixel-space diffusion models. All results are reported with JiTB/16 (Li & He, 2025b), batch size 256, and without classifier-free guidance. See Table 10 for further results.

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on different components.** Across every encoder, both the spatial normalization layer and the convolutional projection layer (§4) yield substantial gains in convergence speed; the two are complementary, and combining them gives the best results. All results reported at 100K steps with SiT-XL/2.

(22.2% for PE-B, 38.8% for PE-L, and 39.6% for PE-G)—precisely the regime in which larger encoders tend to lose spatial structure, and thus the regime in which restoring it is most valuable.

Scalability. Table 1b examines how iREPA scales with the size of the diffusion model. The spatial modifications not only improve performance at every model scale, but yield progressively larger relative gains as the model grows, demonstrating that the benefit of accentuating spatial transfer scales favorably with model size rather than diminishing.

Encoder depth. Table 1c reports iREPA across alignment depths, with all results obtained on SiT-B/2 at 100K iterations using DINOv3-B as the target representation. iREPA improves consistently over baseline REPA at every alignment depth, confirming that the gains are not tied to a particular choice of alignment layer.

Ablation on different components. We isolate the contribution of each iREPA component in Table 2. Both the spatial normalization layer and the convolutional projection layer independently improve generation quality over baseline REPA, and the two are complementary: applying them jointly yields the best results, confirming that each addresses a distinct source of spatial information loss.

Training recipe. Finally, we test whether iREPA transfers across representation alignment recipes, evaluating it on REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025). As Table 3 shows, the spatial modifications deliver convergence gains under both recipes, indicating that accentuating spatial transfer is a recipe-agnostic principle rather than an artifact of the original REPA setup.

Classifier-free guidance. Table 4 reports generation quality with classifier-free guidance across vision encoders at 400K training iterations. iREPA accelerates convergence for every encoder both with and without classifier-free guidance, confirming that its gains persist under the guidance regime used at inference.

Pixel-space diffusion (JiT). We further extend iREPA to pixel-space diffusion models, evaluating it on JiT-B (Li & He, 2025b). Figure 9 contrasts REPA and iREPA under JiT (Li & He, 2025b): accentuating the transfer of spatial information again yields consistently faster convergence across

Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DiNOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DiNOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DiNOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DiNOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** Applied on top of REPA-E (a) and MeanFlow with REPA (b), our spatial modifications yield consistent gains under both recipes. For MeanFlow, we report results at 1 and 4 NFEs, using a CFG value of 2.0. All results are reported at 100K training iterations.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DiNOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+iREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DiNOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+iREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DiNOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+iREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DiNOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+iREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+iREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+iREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+iREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+iREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across encoders, with and without classifier-free guidance.** Using SiT-XL/2 as the base model, iREPA improves generation quality for every encoder under both guidance settings and at both 100K and 400K steps. See Appendix E for detailed results on SiT-B/2 & SiT-L/2 across all encoders.

vision encoders (e.g., DiNOv2, DiNOv3, PE *etc.*), demonstrating that the principle carries over from latent-space to pixel-space diffusion.

- **Finding 3.** *Accentuating transfer of spatial structure from target representation to diffusion features helps boost convergence speed with representation alignment (REPA).*

5 RELATED WORK

We situate our contribution among the most directly relevant lines of work here, and defer an extended discussion to Appendix H.

Representation alignment for generation. A growing body of work exploits external representations to improve the training of diffusion models (Pernias et al., 2023; Fuest et al., 2024). In particular, (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) demonstrate that aligning the internal features of a diffusion model with the clean image features of a pretrained vision encoder yields substantial performance gains, and this idea has since been extended to video generation (Zhang et al., 2025) and 3D generation (Wu et al., 2025), as well as to the training of unified models (Ma et al., 2025a). Despite these empirical successes, the precise mechanism by which self-supervised features enhance diffusion training has remained poorly understood. We close this gap by identifying which property of the target representation actually drives generation—its spatial structure—and by translating that understanding into an improved training recipe.

Spatial vs. global information tradeoff in pretrained vision encoders. A parallel line of work investigates the tension between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) show that continued self-supervised training can increase the similarity between the global CLS token and the patch tokens, degrading performance on dense spatial tasks, while (Bolya et al., 2025; Heinrich et al., 2024) explicitly train spatially tuned models for such tasks. Building on this tradeoff, we establish that for generation it is the spatial structure of a vision encoder, rather than its global information, that is decisive—a finding we hope will reorient how external representations are selected and trained for generative modeling.

6 CONCLUSION

This paper resolves a fundamental question about representation alignment: what property of the target representation actually drives its effectiveness—global information or spatial structure? Through a large-scale empirical analysis spanning 27 vision encoders and three model scales, we establish that it is spatial structure, and not global information, that governs the effectiveness of representation alignment. We then operationalize this principle with two minimal modifications that accentuate the transfer of spatial information from the target representation to the diffusion features. The resulting method, iREPA, consistently accelerates the convergence of REPA across diverse vision encoders and training recipes. By reframing the working mechanism of representation alignment, our findings chart a concrete path toward selecting and training external representations more effectively for generative modeling.

REPRODUCIBILITY STATEMENT

We provide all implementation details and hyperparameters in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

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A DETAILED CORRELATION ANALYSIS

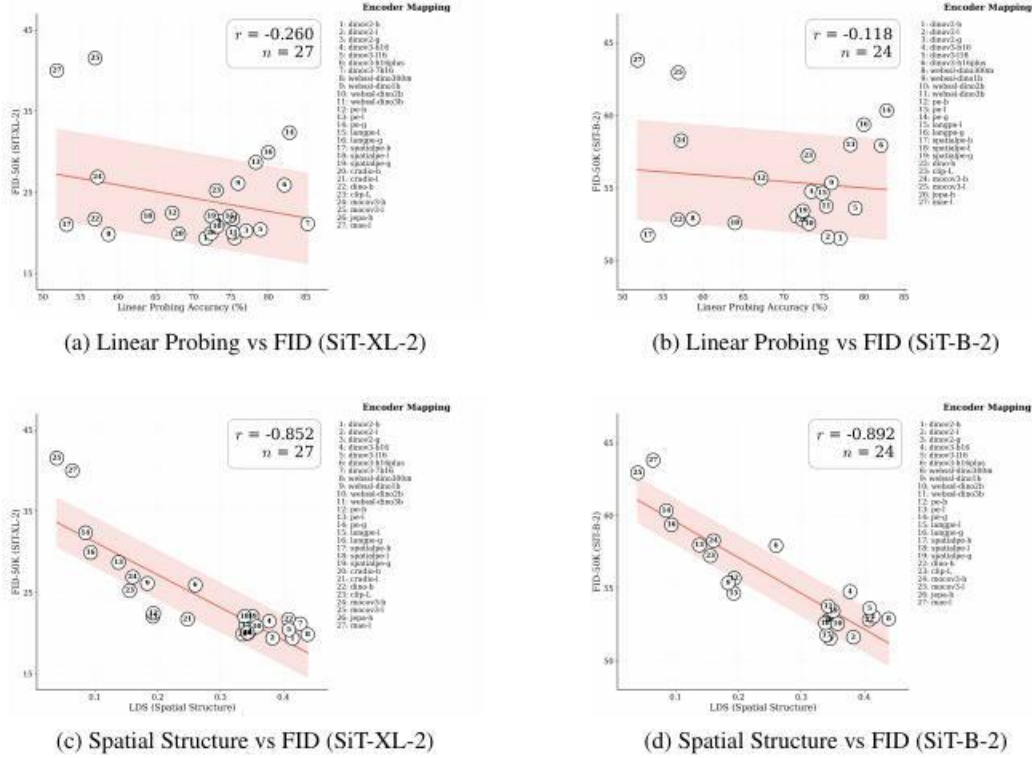


Figure 10: Detailed correlation analysis with encoder labels. The top row plots linear probing accuracy against FID for the SiT-XL-2 and SiT-B-2 models, and the bottom row plots spatial structure (LDS) against FID; each point is labeled with its encoder number (see legend). At both model scales, the spatial structure metric correlates substantially more strongly with generation quality than linear probing accuracy does.

B SPATIAL SELF-SIMILARITY METRICS



Figure 11: **Explaining semantic region self-similarity (SRSS) metric.** Visual explanation of SRSS metric: For each image, we compute a mask using SAM2 and select anchor-positive-negative triplets. The SSM measures the difference in cosine similarity between anchor-positive pairs (within mask) and anchor-negative pairs (outside mask). Intuitively, larger values indicate that patches on the same semantic region are more similar than patches on unrelated regions — indicating better spatial structure preservation.

Setup. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ be the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Intuitively, larger values indicate stronger spatial organization (near patches more similar to each other than far away patches).

We now define several straightforward metrics that quantify the spatial self-similarity structure of the target representations.

Local vs. Distant Similarity (LDS). We first adopt a simple correlogram-contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where r_{near} and r_{far} are hyperparameters, set by default to $r_{\text{near}} = r_{\text{far}} = H/2$; the correlation is robust to their exact values. Larger LDS values indicate that, *on average*, spatially closer patches are more similar than distant ones—a direct signature of preserved spatial structure.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values correspond to faster decay of similarity with distance, and thus to stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ enclosing a random object using SAM2 (Ravi et al., 2024), downsampled to $H \times W$. From this mask we draw three types of points—anchor, positive, and negative—where the anchor and positive points lie within the mask (and are thus semantically related) and the negative point lies outside it (and is thus less related to the anchor). If the encoder preserves spatial structure, the anchor and positive should exhibit high cosine similarity while the anchor and negative should exhibit low similarity. We therefore define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

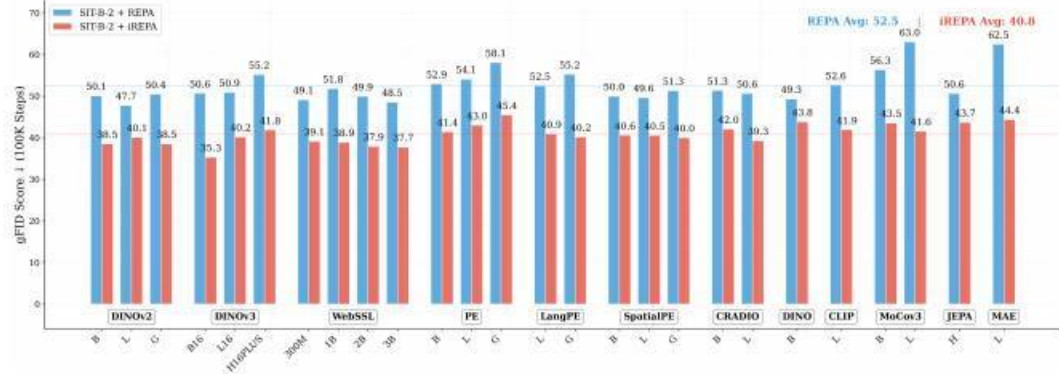
where positive points are within Manhattan distance $d \leq r_{near}$ from the anchor within the mask, and negative points are at distance $d \geq r_{far}$ outside the mask.

RMS Spatial Contrast (RMS C). Finally, we introduce a simple contrast metric that captures the dispersion among patch token representations. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

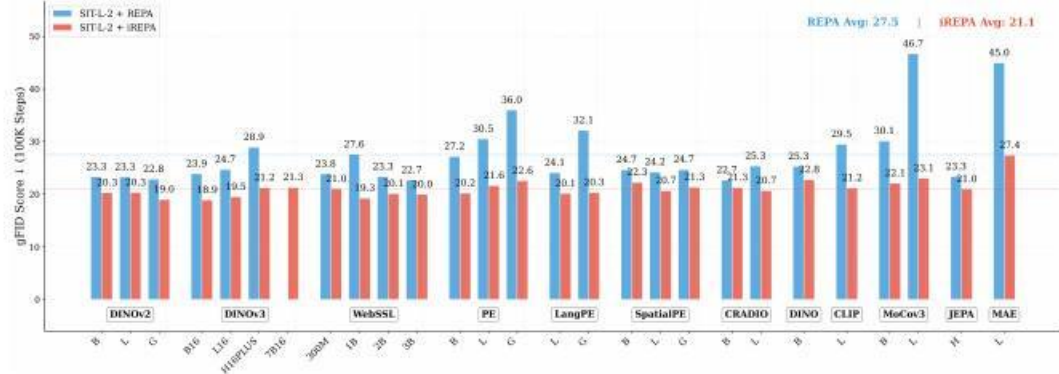
$$\text{RMS C}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean normalized feature across all spatial locations. Higher RMS C values reflect greater spatial diversity among features, and hence preserved spatial structure, whereas lower values reflect a more uniform feature distribution—the hallmark of a loss of spatial structure that arises when a dominant global component overwhelms it (Siméoni et al., 2025).

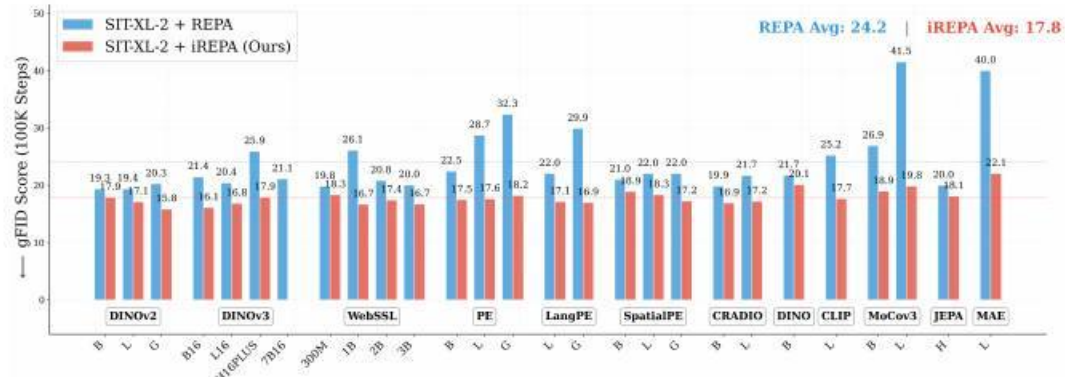
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



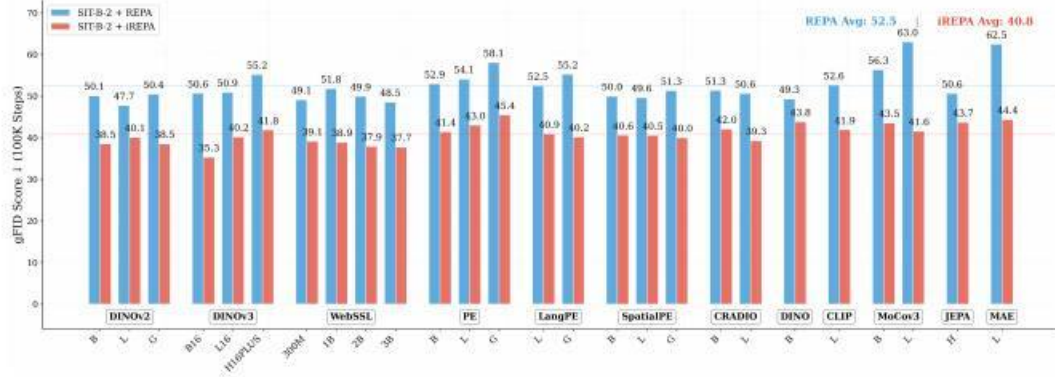
(b) SiT-L-2 (458M parameters)



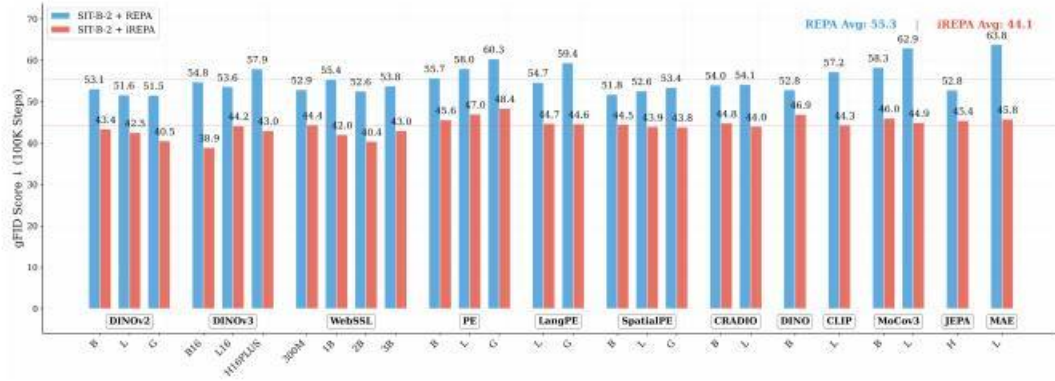
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation studies showing iREPA improvements across different vision encoders for SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA consistently improves generation quality across all encoders and model sizes.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** SiT-B-2 performance at two encoder depths (6 vs. 8 layers) across a range of vision encoders. iREPA improves consistently regardless of encoder depth—slightly more so at 8 layers—and both configurations exhibit substantial FID gains once spatial regularization is applied, confirming that the benefit is not tied to a particular depth.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA improves generation quality for every vision encoder considered. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA improves generation quality for every encoder, with especially pronounced gains in FID and IS. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

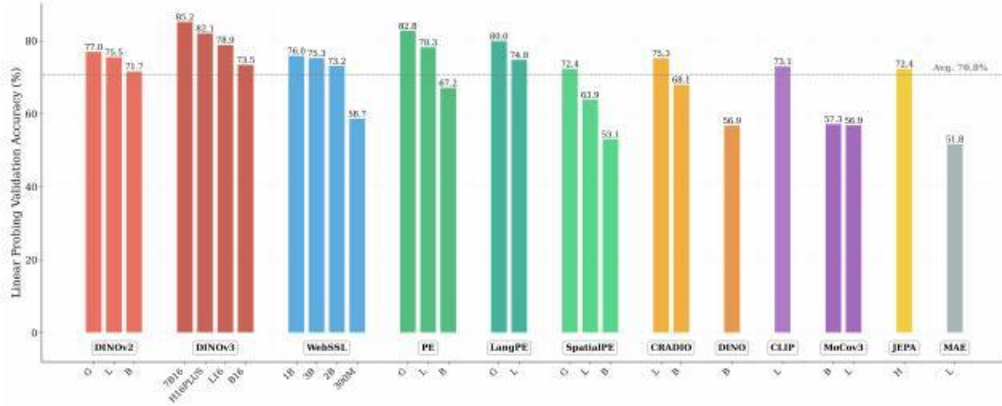


Figure 14: **Global information in mean patch tokens.** We find that in addition to [CLS] token, mean of patch tokens also contains substantial global semantic information. While this helps improve global performance, it reduces the contrast between individual patch tokens, potentially hindering spatial structure transfer. We find that we can remove some of this global information (through mean of patch tokens) to improve spatial structure transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** Comparison of REPA vs iREPA with CFG (scale 2.0) across different vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA consistently improves both IS and FID metrics across all encoders.

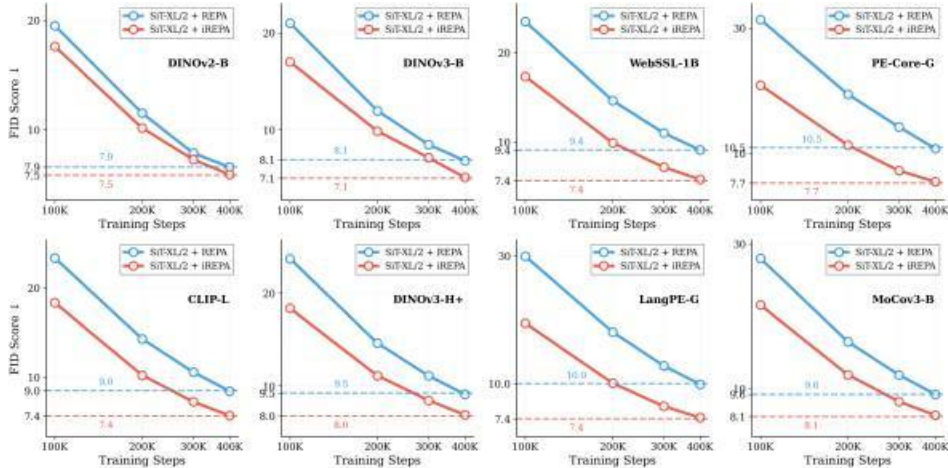


Figure 15: **Accentuating spatial features consistently improves convergence speed.** Convergence results for SiT-XL/2 with REPA and iREPA.

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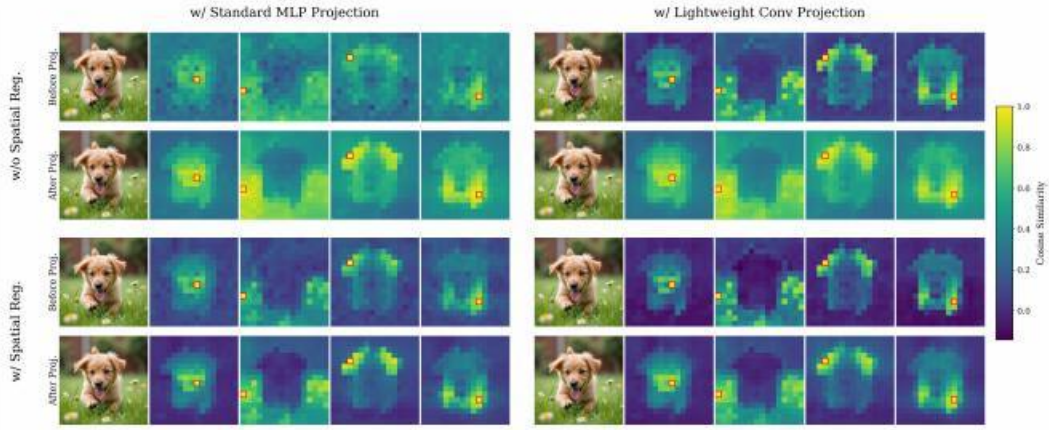


Figure 16: **Visualizing the impact of two straightforward improvements to enhance spatial feature transfer.** **First**, we find that standard MLP projection layer (top-left) losses spatial information while transferring features from the pretrained encoder (after projection) to diffusion model (before projection). Instead using a simpler convolution layer better preserves the spatial information transfer (top-right). **Second**, we observe that vision encoder features often have limited spatial contrast (§2. This causes the tokens in one semantic region (*e.g.*, dog) to show quite decent cosine similarity with unrelated tokens (*e.g.*, background). We address this with a simple spatial regularization layer, which accentuates the spatial contrast in the learned representation — leading to better generation performance. Best results are obtained while using both (refer Table 2).

G IMPLEMENTATION DETAILS

Training setup. Unless otherwise specified, we follow the setup of REPA (Yu et al., 2024). All training is conducted on the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the ADM (Dhariwal & Nichol, 2021) protocol, center-cropping and resizing each image to 256×256 resolution, and we use the `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout diffusion training and inference. The spatial normalization layer uses $\gamma \in [0.6, 0.8]$, and the projection layer is a convolutional layer with kernel size 3 and padding 1. Optimization uses AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} and a global batch size of 256. For efficiency and stability, we train with `bfloat16` mixed precision and `torch.compile`, and apply gradient clipping together with an exponential moving average (EMA) of the generative model.

For the REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we build on the official open-source implementations. For the MeanFlow (Geng et al., 2025) experiments, we obtained the implementation (including REPA) from the original authors and adapted it to incorporate the two minimal iREPA modifications (§4).

Evaluation. For image generation, we follow the ADM evaluation protocol (Dhariwal & Nichol, 2021) exactly. We report generation quality using Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), and precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), all measured over 50K generated images. For sampling, we follow REPA (Yu et al., 2024) and use the SDE Euler–Maruyama sampler with 250 steps; for JiT (Li & He, 2025a) we use 50 inference steps, matching its official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been foundational to computer vision, with classical descriptors such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) providing robust local representations for tasks ranging from object detection to image matching. Although these handcrafted features excel at capturing local spatial patterns and geometric invariances, recent generative modeling has concentrated almost exclusively on modern self-supervised representations that achieve strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings motivate a different perspective: because spatial structure preservation is what governs generation quality, even classical spatial descriptors can improve diffusion training when properly aligned. This points to the value of the full spectrum of spatial feature extractors—from traditional handcrafted descriptors to modern learned representations—provided they maintain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have served generative models in several capacities: as discriminators that accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and, more recently, as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have both improved convergence speed and enabled scaling to large datasets, as shown by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) using CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation exploits pretrained encoders to guide student networks toward higher-fidelity samples, yielding clear perceptual gains. Most directly, REPA (Yu et al., 2024) aligns diffusion features with an external encoder, establishing that representation alignment improves both convergence and generation quality. We build on this direction but shift the emphasis from discriminative capability to spatial structure preservation, which our analysis identifies as the decisive property.

Denosing transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, spanning diffusion transformers (Peebles & Xie, 2023) and flow-matching variants (Ma et al., 2024). Recent architectures such as GenTron (Chen et al., 2023) scale transformers beyond 3B parameters for text-to-image synthesis, U-ViT (Bao et al., 2023) shows that pure transformer backbones without U-Net inductive biases remain competitive, and ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that couple denoising with sequential generation to trade off quality against speed. Despite these advances, training such

models from scratch remains computationally expensive, often demanding millions of iterations to reach good generation quality. Although representation alignment methods like REPA (Yu et al., 2024) have shown that pretrained features can dramatically accelerate convergence, the mechanism behind this acceleration remained unclear. Our analysis resolves it: the benefit stems from preserving spatial structure rather than from semantic alignment, which explains why certain encoders accelerate training more than others and directly informs the design of more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denosing as a self-supervised learning task. The connection between denoising and representation learning has been examined from several angles. Early work by Abstreiter et al. (2021) extended the diffusion objective to representation learning, showing that denoising naturally acquires meaningful features; SODA (Hudson et al., 2024) couples a diffusion model with an information bottleneck to learn compact representations; and Chen et al. (2024) deconstruct diffusion models to show that a simple denoising autoencoder already yields strong self-supervised performance. Wang & He (2025) introduce a dispersive loss that encourages internal representation diversity and improves generation without external encoders, and Jiang et al. (2025) propose Self-Representation Alignment (SRA), which aligns a diffusion transformer’s latent features across noise levels to provide self-guidance without an auxiliary model. Collectively, these works establish denoising as a fundamental self-supervised task that naturally encourages robust features. Our findings complement this view: aligning diffusion models with the spatial representations of self-supervised encoders improves both generative and discriminative capability, identifying spatial structure preservation as the key factor underlying this synergy.

Scaling self-supervised vision encoders. Recent years have brought remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B-parameter ViT on 1.7 billion unlabeled images by aligning representations across augmentations; WebSSL (Fan et al., 2025) shows that models trained on more than 2 billion images can match language-supervised models such as CLIP (Radford et al., 2021) on vision-language tasks without any language supervision; and C-RADIO (Heinrich et al., 2024) distills multiple teachers into versatile encoders that excel across visual domains. I-JEPA (Assran et al., 2023) learns by predicting masked regions in representation space rather than reconstructing pixels, SAM (Ravi et al., 2024) specializes in promptable segmentation via large-scale supervision, and the Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often surpass final representations for dense prediction. These models are almost always evaluated on global tasks such as image classification, yet our work exposes a crucial gap: strong discriminative performance does not translate into better generation. Instead, the encoders that preserve spatial structure—irrespective of their classification accuracy—provide the greatest benefit for diffusion training. This argues that the evaluation criteria for self-supervised encoders should be reconsidered for generative applications, treating spatial coherence as no less important than semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) for considering different variations of the spatial structure metrics discussed in the paper §3. Furthermore, all figures in the paper are directly generated from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). Additionally we use LLM help for searching and formulating relevant work in Appendix H. We use cursor in some parts to help with paper writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can measure the representation gap. Yu et al. (2024) measure the representation gap through linear probing, invoking the resulting increase in validation accuracy to explain the effectiveness of REPA. We show that spatial structure metrics (SSM) can measure this gap equally well and account for improvements across models: as Figure 17 demonstrates, representation alignment closes the gap between the SSM of a pretrained encoder such as DINOv2 and that of the diffusion features.

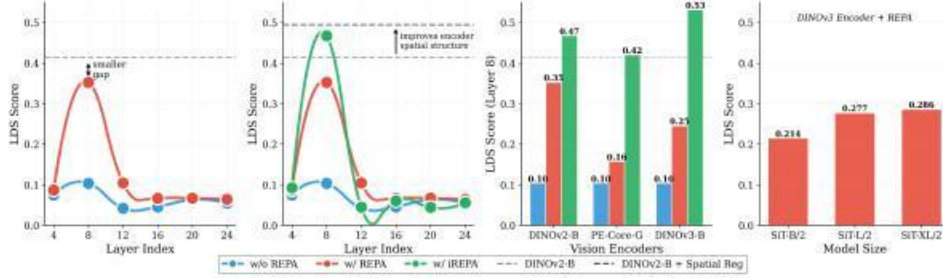


Figure 17: Representation alignment as bridging the spatial feature gap. a) We find that representation alignment can also be seen as bridging the spatial feature gap between diffusion and vision encoder patch features. b) iREPA (§4) accentuates the spatial features of the vision encoder (at cost of some global information) — helping achieve better generation performance. c) iREPA helps consistently improve performance across different vision encoders. d) Spatial structure improvements scale with model size. All results are reported with SiT-XL/2 at 400k iterations.

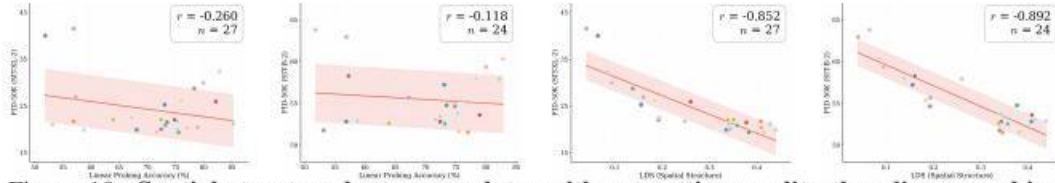


Figure 18: Spatial structure better correlates with generation quality than linear probing. Correlation analysis across 27 vision encoders. **Left two:** Linear probing accuracy vs FID for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$) shows weak correlation. **Right two:** Spatial structure (LDS) vs FID for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$) shows strong correlation. See Figure 10 in Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

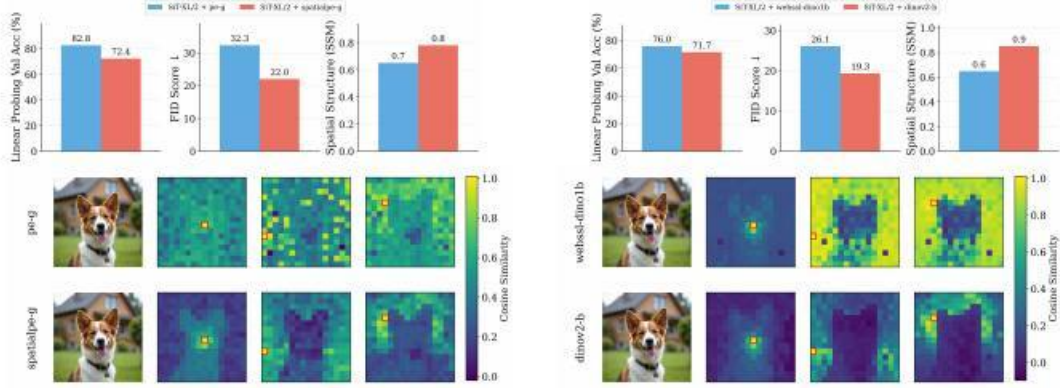


Figure 19: **Motivating anecdotes from recent SSL encoders — spatial structure matters.** **Top row:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g achieves higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, with much lower spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b shows higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine similarity visualizations confirm that encoders with better generation (Spatial-PE-g, DINOv2-b) maintain clear spatial coherence, while those optimized for classification (PE-g, WebSSL) lose spatial structure.

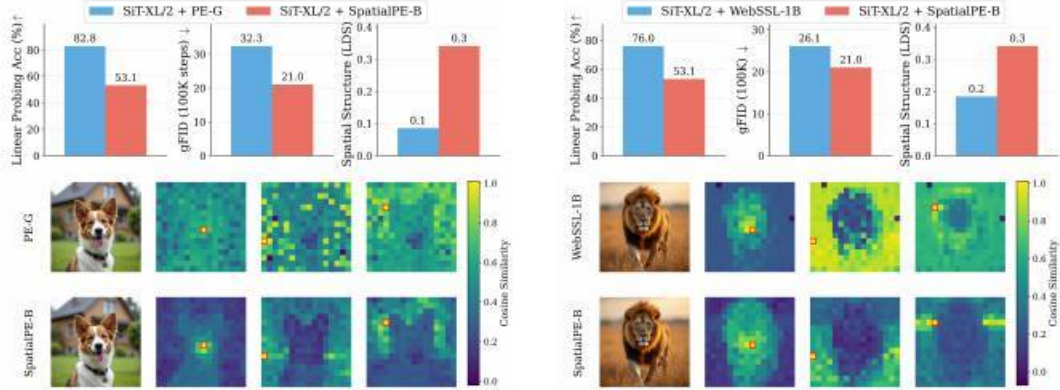


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite having significantly higher validation accuracy (82.8% vs. 53.1%), shows worse generation quality compared to PE_{Spatial}-B (Bolya et al., 2025). **Right:** Similarly, WebSSL-1B (Fan et al., 2025) also shows much better global performance (76.0% vs. 53.1%), but worse generation. **Bottom:** We find that spatial structure instead provides a better predictor of generation quality than global performance. See §3 for spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization. We visualize the effect of spatial normalization on vision encoder representations. Pretrained encoders often carry a substantial global component that suppresses the contrast among spatial tokens, so that tokens in one semantic region become highly correlated with tokens in unrelated regions (e.g., background) and lose their spatial distinctiveness. Spatial normalization removes this global overlay, sharpening the contrast between spatial tokens; the model thereby preserves local spatial structure while shedding the global information that otherwise interferes with generation quality.

The following figures illustrate this effect across several examples, showing how spatial normalization reshapes the feature representations to emphasize spatial structure:

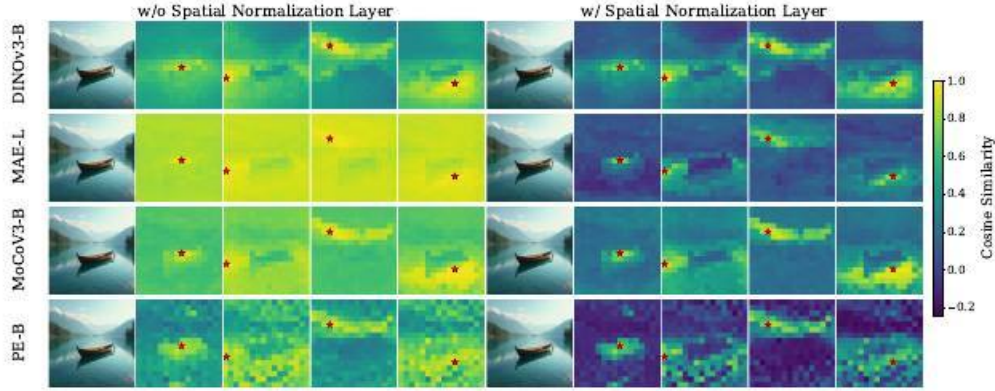


Figure 21: Visualizing impact of spatial normalization (Example 1). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

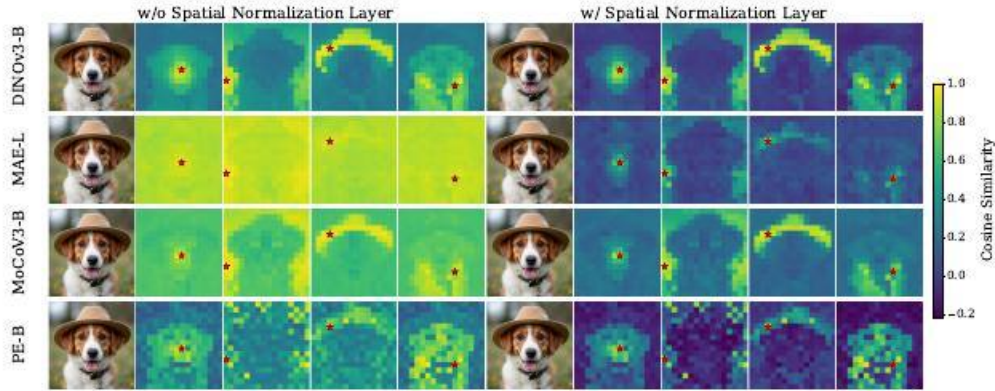


Figure 22: Visualizing impact of spatial normalization (Example 3). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

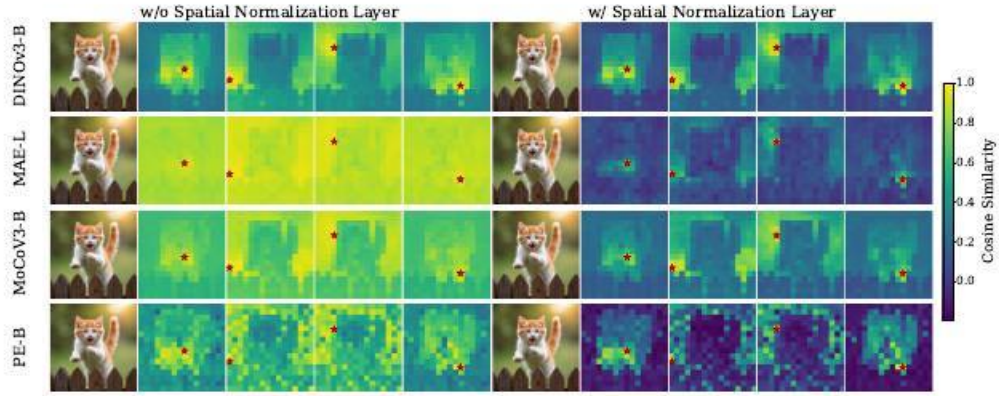


Figure 23: Visualizing impact of spatial normalization (Example 2). Feature similarity maps demonstrate how spatial normalization improves spatial contrast. The original features (left) exhibit a global overlay that reduces distinction between foreground and background regions. After normalization (right), spatial tokens become more locally coherent, with clearer separation between different semantic regions.

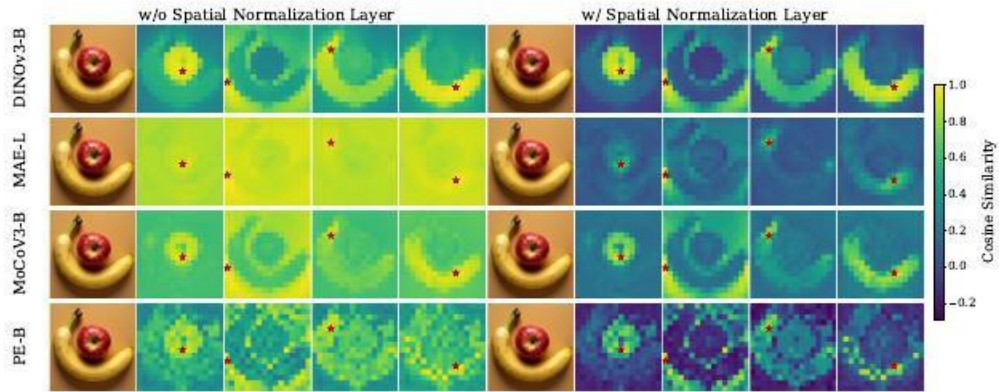


Figure 24: **Visualizing impact of spatial normalization** (Example 4). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We conduct further experiments on ImageNet-512 (Deng et al., 2009) to assess how the proposed spatial improvements generalize to higher resolutions. Following the REPA (Yu et al., 2024) setup, we report results across a range of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.), using SiT-XL and SiT-B with 50 NFE at inference and without classifier-free guidance. Fig. 25 contrasts REPA before and after applying the spatial improvements (iREPA). The spatial improvements generalize cleanly to higher resolution, delivering consistent gains across every pretrained encoder considered (DINOv2, DINOv3, WebSSL, PE, etc.).

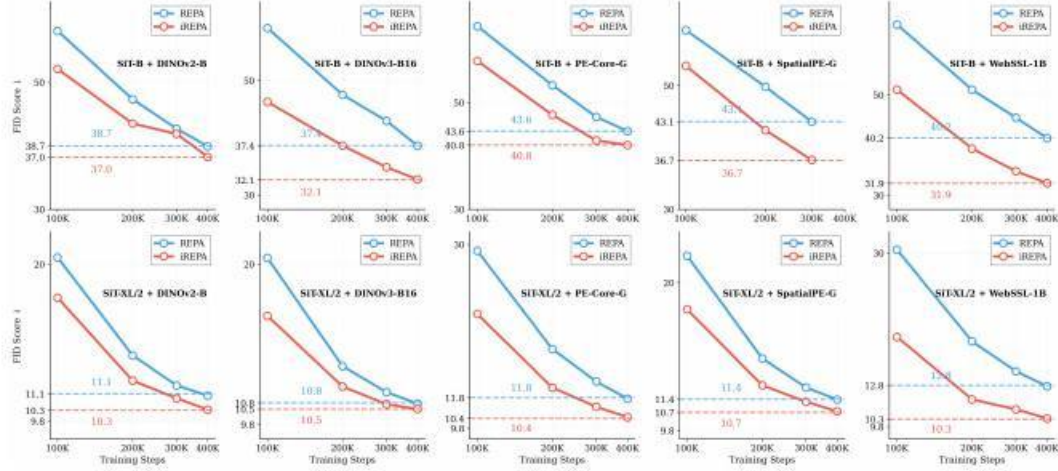


Figure 25: Convergence results at 512x512 resolution. Accentuating spatial features consistently improves convergence speed across resolutions, holding for both ImageNet-256 (Figure 7) and ImageNet-512 (above).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To probe the generalizability of the proposed spatial improvements beyond ImageNet, we further conduct extensive experiments on multimodal text-to-image (T2I) tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with a range of pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). As in REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256 and evaluated with an ODE sampler at 50 NFE. Fig. 26 contrasts REPA before and after applying the spatial improvements (iREPA). The spatial improvements generalize to the multimodal T2I setting as well, yielding consistent gains across every pretrained encoder considered (DINOv2, CLIP, WebSSL, PE, etc.).

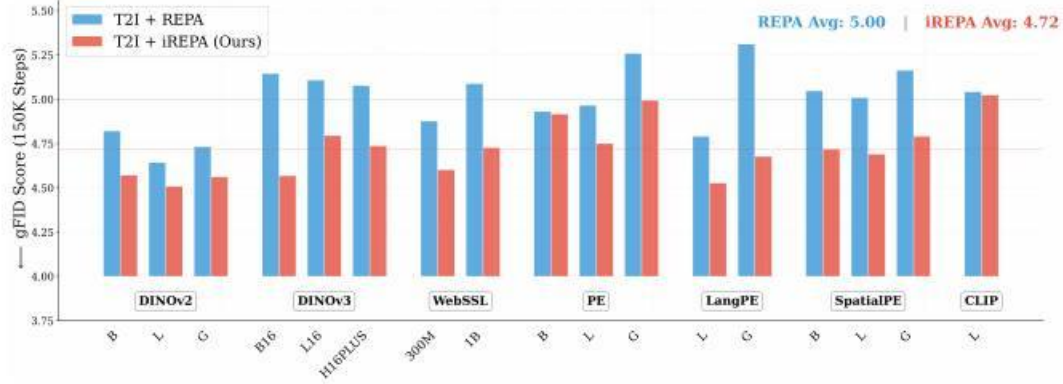


Figure 26: **Text-to-image generation across encoder variants.** Accentuating transfer of spatial features from the target representation to the diffusion features consistently improves convergence speed for both imagenet (refer §4) and multimodal T2I tasks (above). Furthermore, consistent gains are observed across different choice of pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the correlation analysis from Section 3 after removing MoCOv3-L and MAE-L, which were identified as outliers. Figure 28 shows the updated correlations across different model sizes. We observe that spatial structure still shows much higher correlation with generation performance over linear probing accuracy. Interestingly, after removing the outliers linear probing actually shows a small positive correlation with gFID (i.e., as linear probing performance increases, the generation becomes worse). This trend is consistent with the observations discussed in Sec. 2, wherein often target representations with higher global semantic performance (linear probing accuracy) show similar or worse generation performance with representation alignment (REPA).

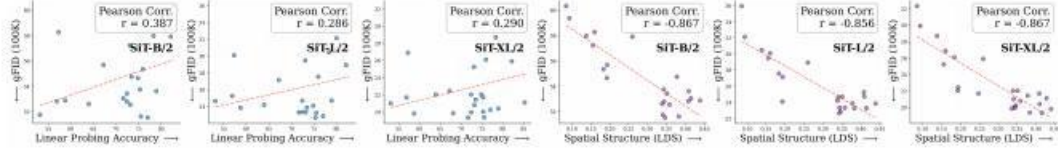


Figure 27: **Correlation analysis without outliers.** Across all model sizes (B, L, XL) spatial structure still shows much higher correlation (Pearson $|r| > 0.85$) with generation performance over linear probing accuracy (Pearson $|r| < 0.38$) after removing the outliers (MoCOv3-L and MAE-L). Interestingly, after removing the outliers linear probing actually shows a small positive correlation with gFID (i.e., as linear probing performance increases, the generation becomes worse). This trend is consistent with the observations discussed in Sec. 2, wherein often target representations with higher global semantic performance (linear probing accuracy) show similar or worse generation performance with representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Additional Qualitative Results** comparing generation outputs before and after application of spatial improvements with REPA. All results are reported using PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs) and with classifier-free guidance scale of 4.0. Similar to quantitative improvements (§4), we also observe that spatial improvements (iREPA) also help improve the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

In addition to traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we also verify the robustness of the proposed findings with alternative evaluation metrics for generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). In particular, we use the CMMD metric (Jayasumana et al., 2024) with the PyTorch implementation from (Paul, 2024). Standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. We next verify the robustness of the key findings from both §3 and §4 with the CMMD metric.

Spatial structure correlates much higher with generation performance than linear probing. To study robustness of analysis from §3, we repeat the large-scale correlation analysis across different vision encoders with CMMD metric (Jayasumana et al., 2024). Results are shown in Figure 29. All results are reported using SiT-B/2 (100K steps) and REPA. All spatial metrics show much higher correlation (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$) — demonstrating that key empirical findings from §3 are robust to the choice of evaluation metric (FID or CMMD).

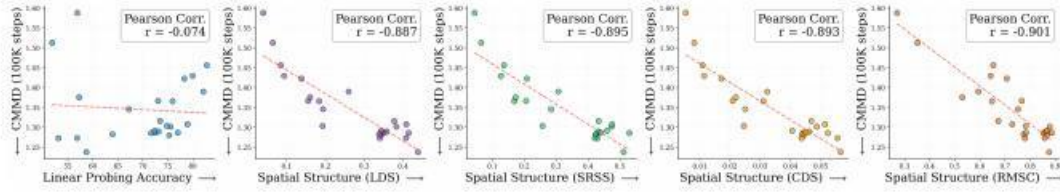


Figure 29: Correlation analysis with CMMD metric. We repeat the correlation analysis from §3 with CMMD metric (Jayasumana et al., 2024) instead of gFID. All results are reported using SiT-B/2 (100K steps) and REPA. All spatial metrics show much higher correlation (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$). This demonstrates that key empirical findings from §3 are robust to the choice of evaluation metric.

Accentuating transfer of spatial features helps improve generation quality. We next study the generalizability of analysis from §4 with the CMMD (Jayasumana et al., 2024) metric in addition to traditional evaluation metrics (gFID, sFID, IS, etc.). Results are shown in Table 8. Across various vision encoders, accentuating transfer of spatial features (iREPA) helps improve convergence speed with both CMMD and traditional evaluation metrics (IS, FID, sFID, Prec., Recall). Furthermore, consistent improvements are observed both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We provide additional results with the CMMD metric (Jayasumana et al., 2024). All results are reported using SiT-XL/2, 256 batch size and traditional REPA as baseline. We adopt the pytorch implementation from (Paul, 2024) for computing the CMMD metric. Standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. Across various vision encoders, accentuating transfer of spatial features (iREPA) helps improve convergence speed with both CMMD and traditional evaluation metrics (IS, FID etc.). This demonstrates that key empirical findings from §4 are robust to the choice of evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 study the impact of spatial normalization (§4) on SAM2-S (46M) encoder features². As shown in Table 9, we observe that for SAM2, while spatial normalization layer helps improve performance, the improvements can be marginal. This is because spatial normalization relies on removing the global component (mean of patch tokens) to enhance spatial contrast. Since SAM2 already has little to no global information (validation accuracy < 24%), spatial normalization only slightly improves the spatial contrast (see Figure 30).

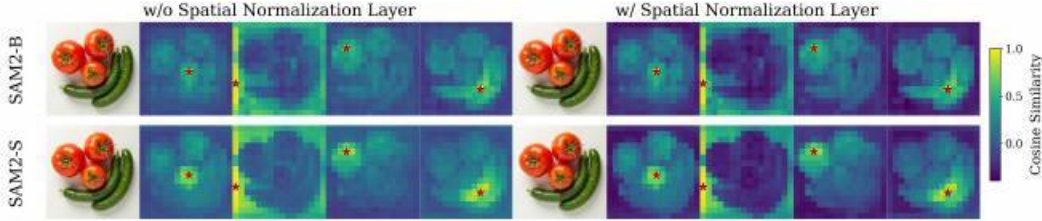


Figure 30: **Impact of spatial normalization on SAM2.** We observe that for SAM2, while use of spatial normalization layer does help enhance the spatial contrast, the improvements can be marginal. This is because spatial normalization (§4) relies on removing the global component (mean of patch tokens) to enhance spatial contrast. Since SAM2 already has little to no global information (validation accuracy < 24%), spatial normalization only slightly improves the spatial contrast.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** All results are reported with SiT-XL/2, without classifier-free guidance, SAM2-S (46M) as vision encoder and with traditional REPA as the baseline.

²Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 with the validation accuracy after full-finetuning³ instead of linear probing. Results are shown in Fig. 31. All results are reported using SiT-XL/2 (100K steps) and REPA. For full-finetuning validation accuracy, instead of linear probing, we perform full-finetuning of vision encoder with learning rate of $5e-5$, warmup ratio of 0.1, and total of 30 epochs. All spatial metrics show much higher correlation with generation performance than full-finetuning accuracy. Interestingly, gFID actually shows a weak positive correlation with the validation accuracy after full-finetuning (Pearson $r = 0.317$), i.e., as validation accuracy increases, the gfid increases, and generation performance becomes worse. This is consistent with the observations discussed in §2, wherein often target representations with higher global semantic performance show similar or worse generation performance with representation alignment (REPA).

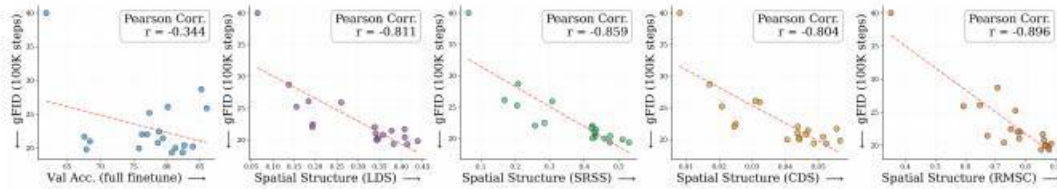


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 with the validation accuracy after full-finetuning instead of linear probing. All results are reported using SiT-XL/2 (100K steps) and REPA. All spatial metrics show much higher correlation with generation performance than full-finetuning accuracy. Interestingly, gFID actually shows a weak positive correlation with the validation accuracy after full-finetuning (Pearson $r = 0.317$), i.e., as validation accuracy increases, the gfid increases, and generation performance becomes worse. This is consistent with the observations discussed in §2, wherein often target representations with higher global semantic performance show similar or worse generation performance with representation alignment (REPA).

³Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Despite its simplicity, when combined with JiT (Li & He, 2025a) iREPA outperforms recently proposed state-of-art pixel-space diffusion methods e.g., DeCo (Ma et al., 2025b). All results are reported with 200K training iterations with a batch size of 256 and evaluated using 50-step Euler sampling without classifier-free guidance. Results for DeCo are obtained directly from (Ma et al., 2025b).